



5 User Authentication Methods that Can Prevent the Next Breach

Gilad David Maayan

There is a growing demand for different types of user authentication technologies for both online and in physical systems. The motivation to authenticate users ranges from access control reasons to business development purposes like adding e-commerce elements.

Organizations need to understand that passwords are not the only way to authenticate users. There is a wide variety of authentication technologies and an even greater range of activities that require authentication methods.

What Is User Authentication?

Authentication is the process of identifying users that request access to a system, network, or device. Access control often determines user identity according to credentials like username and password. Other authentication technologies like biometrics and authentication apps are also used to authenticate user identity.

Why Is User Authentication Important?

User authentication is a method that keeps unauthorized users from accessing sensitive information. For example, User A only has access to relevant information and cannot see the

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Hackers gained access to Yahoo user accounts to steal contacts, calendars and private emails between 2012 and 2016. The [Equifax data breach](#) in 2017 exposed credit card data of more than 147 million consumers. Without a secure authentication process, any organization could be at risk.

5 Common Authentication Types

Cybercriminals always improve their attacks. As a result, security teams are facing plenty of authentication-related challenges. This is why companies are starting to implement more sophisticated [incident response](#) strategies, including authentication as part of the process. The list below reviews some common authentication methods used to secure modern systems.

1. Password-based authentication

Passwords are the most common methods of authentication. Passwords can be in the form of a string of letters, numbers, or special characters. To protect yourself you need to create strong passwords that include a combination of all possible options.

However, passwords are prone to [phishing](#) attacks and bad hygiene that weakens effectiveness. An average person has about 25 different online accounts, but only [54%](#) of users use different passwords across their accounts.

The truth is that there are a lot of passwords to remember. As a result, many people choose convenience over security. Most people use simple passwords instead of creating reliable passwords because they are easier to remember.

The bottom line is that passwords have a lot of weaknesses and are not sufficient in protecting online information. Hackers can easily guess user credentials by running through all possible combinations until they find a match.

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ways to identify a user. Examples include codes generated from the user's smartphone, Captcha tests, fingerprints, voice biometrics or facial recognition.

MFA authentication methods and technologies increase the confidence of users by adding multiple layers of security. MFA may be a good defense against most account hacks, but it has its own pitfalls. People may lose their phones or SIM cards and not be able to generate an authentication code.



3. Certificate-based authentication

Certificate-based authentication technologies identify users, machines or devices by using digital certificates. A digital certificate is an electronic document based on the idea of a driver's license or a passport.

The certificate contains the digital identity of a user including a public key, and the digital signature of a certification authority. Digital certificates prove the ownership of a public key and issued only by a certification authority.

Users provide their digital certificates when they sign in to a server. The server verifies the credibility of the digital signature and the certificate authority. The server then uses cryptography to confirm that the user has a correct private key associated with the certificate.

4. Biometric authentication

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- You can add biometrics into your multi-factor authentication process.

Biometric authentication technologies are used by consumers, governments and private corporations including airports, military bases, and national borders. The technology is increasingly adopted due to the ability to achieve a high level of security without creating friction for the user. Common biometric authentication methods include:

- **Facial recognition**—matches the different face characteristics of an individual trying to gain access to an approved face stored in a database. Face recognition can be inconsistent when comparing faces at different angles or comparing people who look similar, like close relatives. Facial liveness like ID R&D's [passive facial liveness](#) prevents spoofing.
- **Fingerprint scanners**—match the unique patterns on an individual's fingerprints. Some new versions of fingerprint scanners can even assess the vascular patterns in people's fingers. Fingerprint scanners are currently the most popular biometric technology for everyday consumers, despite their frequent inaccuracies. This popularity can be attributed to iPhones.
- **Speaker Recognition** —also known as voice biometrics, examines a speaker's speech patterns for the formation of specific shapes and sound qualities. A voice-protected device usually relies on standardized words to identify users, just like a password.
- **Eye scanners**—include technologies like iris recognition and retina scanners. Iris scanners project a bright light towards the eye and search for unique patterns in the colored ring around the pupil of the eye. The patterns are then compared to approved information stored in a database. Eye-based authentication may suffer inaccuracies if a person wears glasses or contact lenses.

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5. Token-based authentication

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Conclusion

Authentication technology is always changing. Businesses have to move beyond passwords and think of authentication as a means of enhancing user experience. Authentication methods like biometrics eliminate the need to remember long and complex passwords. As a result of enhanced authentication methods and technologies, attackers will not be able to exploit passwords, and a data breach will be prevented.

If you are looking to incorporate biometrics into your authentication processes, [contact an ID R&D specialist](#).

Want to learn more about how ID R&D enables secure and frictionless biometric login?

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Author Bio

Gilad David Maayan is a technology writer who has worked with over 150 technology companies including SAP, Samsung NEXT, NetApp and Imperva, producing technical and thought leadership content that elucidates technical solutions for developers and IT leadership.

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Facial Liveness Detection: Which is Best?



Passive vs Active Facial Liveness Detection: Which is Best?

Biometrics Part of Your Multi-Factor Authentication Strategy



5 Reasons to use Biometrics for Multi-factor Authentication

WHAM! WONDER TWIN POWERS AUTHENTICATE!

81%

of consumers view biometrics as a MORE
SECURE FORM of identity verification
2020 Experian Global Identity and Fraud Report

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